# Quantitative Analysis and Game Theory in Tee-Taek Card Game Strategies: A Case Study on Risk Management and Probability for Profit Maximization

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## 1 Abstract

"Tee-Taek" is a widely popular traditional card game in Thailand that currently lacks formal analytical study at an international level. While generally perceived as a gambling game reliant on luck, its mathematical structure allows for the profound application of probability theory, portfolio management, and behavioral psychology. This research aims to examine and compare the efficiency of two primary strategies: the Light Strategy, which relies on decision-making based on visible card data, and the Dark Strategy, which leverages the asymmetric risk inherent in the game's rules. The ultimate objective is to maximize the expected value, aligning the gameplay with professional poker methodologies. Through a Monte Carlo simulation of 1,000,000 rounds, this paper demonstrates the disparity in average returns between the two strategies. It also analyzes the recommended bankroll and emotional management required under volatile conditions to maintain a statistical edge within a limited timeframe. The findings indicate that a systematic understanding of the underlying mathematics and risk management is the critical factor that transforms this game of chance into a strategic investment.

## 2 Introduction

Within the context of Thai recreational culture, "Tee-Taek" is deeply rooted in society. However, from the perspectives of statistics and game theory, the game possesses a mathematical complexity that transcends mere gambling. Tee-Taek serves as a perfect model for decision-making under uncertainty, where players

must navigate risk and probability in a manner akin to international strategy games like Poker.

## 2.1 Research Objectives and Hypotheses

The primary goal of this research is to discover a strategy capable of generating profit maximization under various constraints. The core hypothesis posits that the statistical edge does not solely depend on visible information but may emerge from an asymmetric payoff structure. Furthermore, the research focuses on calculating the initial capital requirements necessary to withstand long-term game volatility.

## 2.2 Game Mechanics

To ensure quantitative accuracy, this research establishes a standard rule structure for calculations. First, regarding components and initiation, the game accommodates 4 to 7 players, each contributing a seed money of 200 THB to the central pot. Every player receives 4 cards from a standard 52-card deck. Second, the game utilizes a circular hierarchy. The cards consist of 4 suits and 13 ranks (2-10, J, Q, K, A). A special rule dictates that a "2 beats an A" and an "A loses only to a 2.". This circular ranking makes the probability calculations distinctly different from linear-ranked card games. Finally, the winning conditions are evaluated after players finalize their strategy. A center card is revealed, and a player wins only if their hand meets two criteria: it must contain at least one card of the same suit as the center card, and that specific card must possess a higher rank than the center card.

## 2.3 Structural Comparison: Light Strategy vs. Dark Strategy

The theoretical highlight of this research is the clash between two strategies with entirely different risk profiles. The Light Strategy involves decision-making with full information. Players inspect their cards before placing a bet, allowing them to engage only during highly advantageous situations. However, they bear the risk of a full loss if defeated. Conversely, the Dark Strategy involves decision-making with no information. Players bet without viewing their cards, but receive a structural handicap: a 50% loss protection. If they lose, they only forfeit half their bet, whereas a victory still yields the full reward.

## 2.4 Research Problem and Statistical Methodology

Initial human intuition generally considers the Dark Strategy to be excessively volatile, as betting without viewing one's hand appears irrational. The Light Strategy, which involves selective engagement based on strong hands, is commonly viewed as the safer and more sustainable approach. However, in actuarial

science, the value of the 50% loss protection could be the variable that completely shifts the advantage. This paper employs a Monte Carlo simulation, running tens of thousands of iterations, to determine whether human intuition or the statistics of asymmetric risk leads to maximum long-term profitability.

# 3 Literature Review

Quantitative analysis of Tee-Taek strategies requires the application of fundamental game theory and statistics to explain player behavior and value transfer mechanisms.

## 3.1 Game Theory and Imperfect Information

Tee-Taek is classified as an incomplete/imperfect information game because players receive unequal data. They possess private information regarding their own hands and public information regarding the center card, but they cannot discern their opponents' hands. This condition forces decisions to be made based on probability calculations and expected value assessment to manage risk under restricted information.

## 3.2 Zero-Sum Game and Resource Allocation

By definition, this is a zero-sum game, meaning the net utility of the system equals zero. The profit gained by one player inherently stems from the combined losses of others. The central pot acts as a limited resource. If a player utilizes an algorithm or strategy that yields a statistical edge, allowing them to extract more than the average share from the pot, players employing suboptimal strategies will absorb the losses and the refill fees necessary to maintain the system's cash flow equilibrium.

## 3.3 Dark Strategy and Asymmetric Risk

Simulations reveal that the Dark Strategy generates a significantly positive expected value due to the asymmetric risk structure, specifically the "lose half, win full" payoff. According to the law of large numbers, as the number of trials approaches infinity without liquidity constraints, this strategy yields a guaranteed profit. The statistical expected value is higher than that of the Light Strategy, which is hindered by emotional limitations and overly strict decision criteria that result in high opportunity costs.

## 3.4 Nash Equilibrium and Strategy Decay

A Nash Equilibrium occurs when all players choose their optimal strategy considering the strategies of others, with no individual benefiting from changing their strategy alone. If this research becomes common knowledge and all players adopt the Dark Strategy, an all-dark equilibrium will form. The statistical

edge will instantly degrade, plunging the system into an equilibrium where no player holds an advantage. Capital will merely circulate and continuously diminish due to pot refill fees. Under these conditions, the game devolves from a strategic endeavor into pure gambling, as there are no longer information gaps or opponent errors to exploit for profit.

# 4 Methodology

To test the hypotheses and identify the optimal strategy, this research began with empirical observation from over 100 hours of actual gameplay. It was observed that most players exhibit risk aversion, predominantly utilizing the Light Strategy and engaging only when possessing a highly advantageous hand, such as holding 3 to 4 suits with high-ranking cards (J, Q, K, A). The psychological limitation of this approach is that when the central pot grows large, players fear losing substantial amounts, causing their play frequency to drop to merely 6-8% of total rounds, despite having a high win rate. In contrast, the Dark Strategy forces players to bet in every round (100% frequency), which, despite a lower win rate, is shielded by the 50% loss protection rule.

## 4.1 Expected Value Analysis

To mathematically prove the statistical advantage, the expected value equation for the Dark Strategy was formulated. Following the Monte Carlo simulation, it was determined that the win rate for the Dark Strategy is 38%. When written as a general equation to find the relationship with the bet size, it results in the following:

$$E[X] = (P_{win} \times \text{Bet Size}) + (P_{loss} \times \frac{-\text{Bet Size}}{2})$$

$$E[X] = (0.38 \times \text{Bet Size}) + (0.62 \times \frac{-\text{Bet Size}}{2})$$

$$E[X] = 0.07 \times \text{Bet Size}$$

The result $E[X] = 0.07 \times \text{Bet Size}$ demonstrates a positive coefficient. This indicates that for every unit of currency wagered using the Dark Strategy, the expected net profit is 7% of that wager. Therefore, strategically, if the number of plays approaches infinity, players should choose the maximum possible bet size within their capital constraints to expand the expected value proportionally to the bet.

## 4.2 Simulation Design

A Python program was developed to execute a Monte Carlo simulation of up to 1,000,000 rounds. The system consisted of 6 players starting with a 1,200 THB pot (200 THB per player). The players were divided into 3 Dark Strategy users and 3 Light Strategy users (simulating the behavior of waiting for premium

cards). Card distribution utilized a uniform probability distribution from a standard deck. Initial simulations revealed that the primary variable causing losses for the Light Strategy was the refill burden. Because Light players engaged in only 7.2% of the rounds but had to contribute the 200 THB refill fee every time the pot emptied, their long-term cash flow became negative.

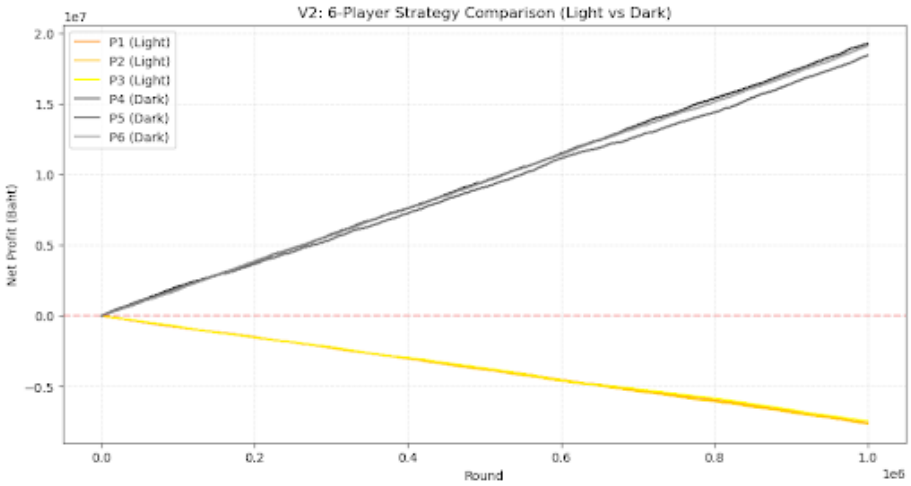


Figure 1: 6-Players Strategy Comparison (Light vs Dark)

# 5 Results and Analysis

Recognizing real-world constraints on time and capital, the simulation was adjusted to reflect realistic environments. A single session was defined as 5 hours of gameplay (approximately 60 rounds), and simulations were repeated to determine statistical deviations.

## 5.1 Case Study 1: The 4-Card Constraint

The simulation of 1,000,000 sessions under the 4-card rule (Dark Strategy win rate of 38%) yielded a daily average profit of +400.36 THB with a standard deviation of 2,437.91 THB. The probability of a session loss was 43.54%. The 95% confidence interval for the worst-case scenario was -4,377.94 THB, while the 3-sigma extreme case was -6,913.36 THB. The maximum observed loss was -10,750 THB, and the maximum observed profit was +12,200 THB. The profit distribution forms a perfect normal distribution. Although the average profit is relatively modest, players must prepare a minimum bankroll of 10,750 THB to prevent the risk of ruin on highly unlucky days falling within the 1% threshold.

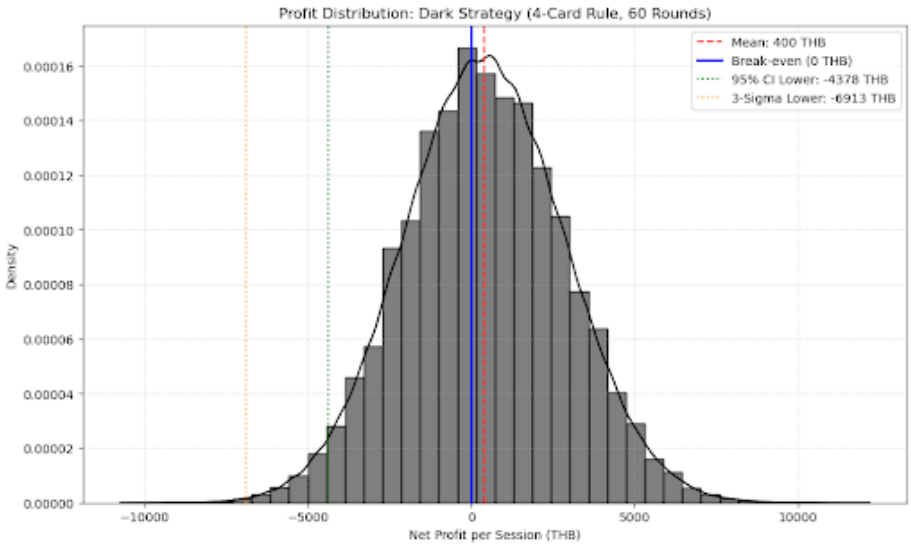


Figure 2: Profit distribution: Dark strategy (4 cards, 60 rounds)

## 5.2 Case Study 2: The 5-Card Asymmetric Advantage

Certain playing environments implement a 5-card rule, where Light players must discard one card, while Dark players utilize all five cards for outcome evaluation. This increase in card volume significantly boosts the Dark Strategy win rate from 38% to over 42%. A 100,000-session simulation under this rule demonstrated an exponential improvement. The average profit per day surged to +2,504.55 THB (a sixfold increase) with a standard deviation of 2,538.44 THB. The probability of a session loss decreased substantially to 15.74%. The 3-sigma extreme case was measured at -5,110.76 THB, with a maximum observed loss of -7,200 THB. The 5-card rule creates an environment where the Dark Strategy entirely dominates the game. The average profit rises significantly while the daily loss risk plummets. Players can safely reduce their required bankroll to 7,500 THB and still comfortably survive in the long term.

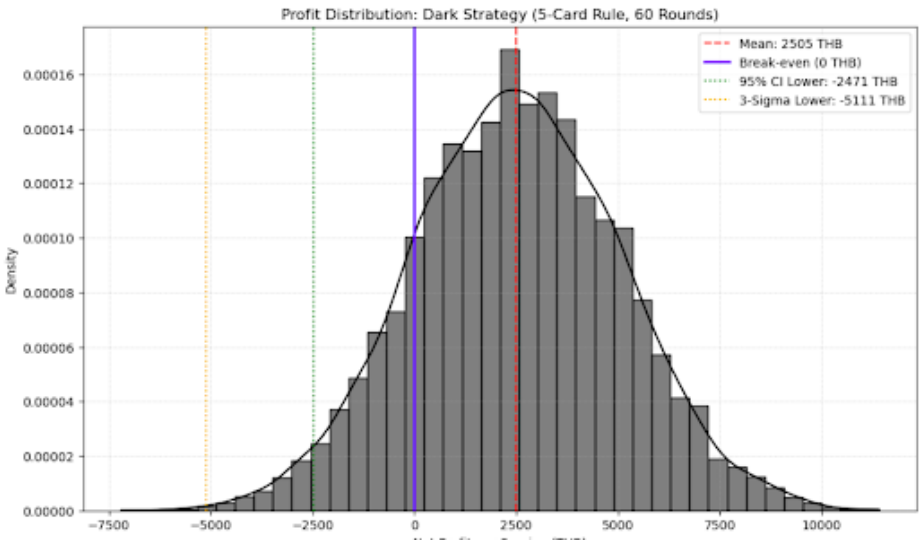


Figure 3: Profit distribution: Dark strategy (5 cards, 60 rounds)

# 6 Limitations and Future Work

While the findings confirm the superiority of the Dark Strategy, practical implementations face certain constraints. First, some betting circles do not permit the Dark Strategy rule. Second, highly restricted time limits or round counts may expose players to short-term variance, preventing them from reaching the necessary volume for the $EV$ to materialize. Consequently, exploring an optimal Light Strategy calculated via the Kelly Criterion for edge-based bet sizing is a critical topic for future research when the Dark Strategy is unfeasible or when volatility risks exceed a player's threshold.

# 7 Conclusion and Recommendations

This research establishes a statistically significant divergence between human intuition and mathematical execution in the Tee-Taek card game. Following the simulation of 1,000,000 rounds, the strategic conclusions and recommendations are detailed below.

## 7.1 Strategic Conclusion

Mathematics consistently triumphs over intuition. Even though the Dark Strategy has a seemingly low win rate of 38% compared to selectively playing premium cards, its long-term expected value remains decidedly positive. The asymmetric payoff of losing half while winning the full amount, combined with a 100%

play frequency, empowers the Dark Strategy to extract significantly more cash from the central pot than the refill fees that burden Light Strategy players.

### 7.2 Bankroll Management

A statistical edge is only functional if a player possesses sufficient capital to survive short-term variance and reach the long term. Based on 3-sigma risk levels, the recommended minimum bankroll for the 4-card rule (yielding a 38% win rate) is 11,000 THB to accommodate high volatility. For the 5-card rule (yielding a 42% win rate), a minimum bankroll of 7,500 THB is recommended, as the volatility shifts to a medium level.

| **Game Rule** | **Win Rate** | **Recommended Bankroll** | **Profit Volatility** |
|---|---|---|---|
| 4-Card Rule | ∼38% | 11,000 THB | High Volatility |
| 5-Card Rule | ∼42% | 7,500 THB | Medium Volatility |

Table 1: Bankroll Management Requirements

### 7.3 Behavioral Insights

Strategic players should actively exploit risk aversion. The Dark Strategy achieves maximum efficiency when deployed amidst Light players who selectively wait for premium hands, as these players act as liquidity providers for the system. Furthermore, players must remain vigilant regarding the Nash Equilibrium. If the mathematical advantage of the Dark Strategy becomes common knowledge, leading all players to adopt it, the statistical edge will be nullified, and the system will revert to pure gambling, where no participant holds an advantage.

### 7.4 Final Remarks

The key to maximizing profit in Tee-Taek is not praying for the best cards, but rather maintaining statistical discipline and systematic risk management. This research illustrates that applying a quantitative approach can successfully transform a game of chance into a highly accurate and sustainable strategic investment.

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